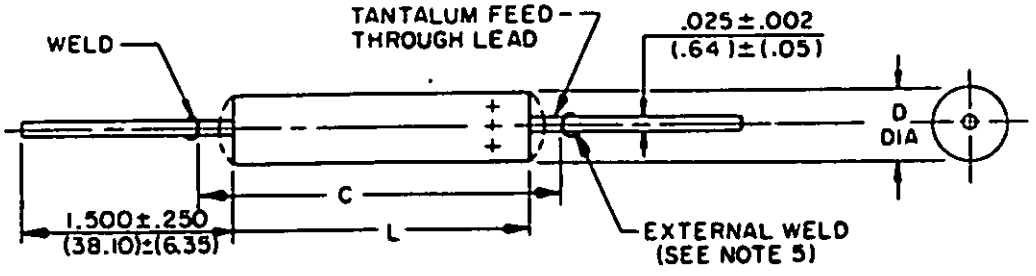
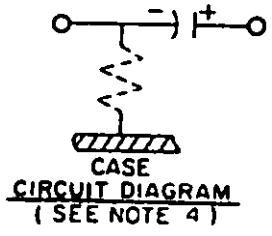
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM, (POLARIZED, HIGH ETCHED FOIL), 85° C (VOLTAGE DERATED TO 125° C), ESTABLISHED RELIABILITY, STYLE CLR71

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of specification MIL-C-39006.





Ⓐ

Case size	Dimensions			
	Basic case		Insulated case	
	L 1/ ±.062 (1.57)	D ±.016 (.41)	D Max	C Max
E1	.688 (17.48)	.188 (4.78)	.219 (5.56)	1.188 (30.18)
E2	.906 (23.01)	.281 (7.14)	.312 (7.92)	1.406 (35.71)
G2	.969 (24.61)	.281 (7.14)	.312 (7.92)	1.406 (35.71)
E3, G3	1.438 (36.53)	.375 (9.53)	.406 (10.31)	1.938 (49.23)
E4, G4	2.125 (53.98)	.375 (9.53)	.406 (10.31)	2.625 (66.68)
E5, G5	2.750 (69.85)	.375 (9.53)	.406 (10.31)	3.250 (82.55)

1/ Length of basic case; sleeving shall be as specified in MIL-C-39006.

NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. There is an indeterminate resistance between the metal case and the negative terminal.
5. The weld shall not be enclosed in the end seal.

FIGURE 1. Style CLR71 capacitors.

REQUIREMENTS:

Design and construction:
Dimensions and configuration - See figure 1.
Case type - Tubular, insulated.
Ⓐ Seal type - G (hermetic) or E (nonhermetic) as shown in table I.
Terminals - Axial-wire lead.
Rated temperature - 55° to +85° C, voltage derated to -125° C (see figure 2).
Rated voltage - See table I.

Design and construction (cont):
Capacitance (Cap.) (nom) - See table I.
Cap. tolerance - See table I.
FR level - See table I.

Constant voltage conditioning: As specified in MIL-C-39006/

DC leakage (DCL) (max) (at 25° C): See table I.

Ⓐ denotes changes

FSC 5910

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:16Z
Check the source to verify that this is the current version before use.

TABLE I. Capacitor characteristics.

Rated voltage (85° C)	Cap.	Cap. tolerance	DC leakage (max)			Derated voltage (125° C)	Surge voltage (85° C)	Impedance (max)	Case size	Part No. M39006/16- Failure rate level (%/1,000 hr)				
			25° C	85° C	125° C					L(2.0)	M(1.0)	P(0.1)	R(0.01)	S(0.001)
volts, dc	μF	%	μA	μA	μA	volts, dc	volts, dc	ohms						
15	29	+75, -15	7	35	23	10	17.2	137	E1	1000	1035	1070	1105	1140
15	100	+75, -15	23	100	80	10	17.2	37	E2	1001	1036	1071	1106	1141
15	310	+75, -15	61	370	250	10	17.2	10.8	E3	1002	1037	1072	1107	1142
15	650	+75, -15	130	780	520	10	17.2	5.1	E4	1003	1038	1073	1108	1143
15	920	+75, -15	185	1100	750	10	17.2	4.0	E5	1004	1039	1074	1109	1144
25	17	+75, -15	6	34	21	15	28.8	220	E1	1005	1040	1075	1110	1145
25	60	+75, -15	23	100	72	15	28.8	62	E2	1006	1041	1076	1111	1146
25	180	+75, -15	60	360	220	15	28.8	20.5	E3	1007	1042	1077	1112	1147
25	380	+75, -15	125	760	460	15	28.8	9.6	E4	1008	1043	1078	1113	1148
25	550	+75, -15	185	1100	660	15	28.8	6.7	E5	1009	1044	1079	1114	1149
30	15	+75, -15	6	34	22	20	34.5	250	E1	1010	1045	1080	1115	1150
30	52	+75, -15	23	120	80	20	34.5	71	E2	1011	1046	1081	1116	1151
30	150	+75, -15	60	360	240	20	34.5	25	E3	1012	1047	1082	1117	1152
30	320	+75, -15	125	750	500	20	34.5	11.5	E4	1013	1048	1083	1118	1153
30	460	+75, -15	165	1000	700	20	34.5	8	E5	1014	1049	1084	1119	1154
50	7	+50, -15	5	28	17	30	57.5	550	E1	1015	1050	1085	1120	1155
50	25	+50, -15	19	100	60	30	57.5	150	E2	1016	1051	1086	1121	1156
50	75	+50, -15	50	300	180	30	57.5	49	E3	1017	1052	1087	1122	1157
50	160	+50, -15	100	640	390	30	57.5	23	E4	1018	1053	1088	1123	1158
50	230	+50, -15	150	920	550	30	57.5	16	E5	1019	1054	1089	1124	1159
75	4	+50, -15	5	24	16	50	86.2	930	E1	1020	1055	1090	1125	1160
75	15	+50, -15	17	90	60	50	86.2	250	E2	1021	1056	1091	1126	1161
75	45	+50, -15	45	270	180	50	86.2	83	E3	1022	1057	1092	1127	1162
75	95	+50, -15	95	570	380	50	86.2	39	E4	1023	1058	1093	1128	1163
75	130	+50, -15	130	780	520	50	86.2	29	E5	1024	1059	1094	1129	1164
100	3	+30, -15	4	20	14	65	115.0	1210	E1	1025	1060	1095	1130	1165
100	10	+30, -15	15	80	52	65	115.0	370	E2	1026	1061	1096	1131	1166
100	30	+30, -15	40	230	150	65	115.0	121	E3	1027	1062	1097	1132	1167
100	63	+30, -15	80	480	310	65	115.0	58	E4	1028	1063	1098	1133	1168
100	90	+30, -15	120	720	470	65	115.0	41	E5	1029	1064	1099	1134	1169
150	1.8	+30, -15	3	18	12	100	172.0	2340	E1	1030	1065	1100	1135	1170
150	5.5	+30, -15	12	64	42	100	172.0	675	E2	1031	1066	1101	1136	1171
150	17	+30, -15	30	190	130	100	172.0	220	E3	1032	1067	1102	1137	1172
150	35	+30, -15	65	400	260	100	172.0	106	E4	1033	1068	1103	1138	1173
150	50	+30, -15	95	560	380	100	172.0	74	E5	1034	1069	1104	1139	1174

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:16Z
Check the source to verify that this is the current version before use.

TABLE I Capacitor characteristics. - Continued

Rated voltage (65° C)	Cap.	Cap. tolerance	DC leakage (max)			Derated voltage (125° C)	Surge voltage (85° C)	Impedance (max)	Case size	Part No. M39006/16- Failure rate level (%/1,000 hr)				
			25° C	85° C	125° C					L(2.0)	M(1.0)	P(0.1)	R(0.01)	S(0.001)
volts, dc	µF	%	µA	µA	µA	volts, dc	volts, dc	ohms						
15	100	+75, -15	23	100	80	10	17.2	37	G2	2001	2036	2071	2106	2141
15	310	+75, -15	61	370	250	10	17.2	10.8	G3	2002	2037	2072	2107	2142
15	650	+75, -15	130	780	520	10	17.2	5.1	G4	2003	2038	2073	2108	2143
15	920	+75, -15	185	1100	750	10	17.2	4.0	G5	2004	2039	2074	2109	2144
25	60	+75, -15	23	100	72	15	28.8	62	G2	2006	2041	2076	2111	2146
25	180	+75, -15	60	360	220	15	28.8	20.5	G3	2007	2042	2077	2112	2147
25	380	+75, -15	125	760	460	15	28.8	9.6	G4	2008	2043	2078	2113	2148
25	550	+75, -15	185	1100	660	15	28 8	6.7	G5	2009	2044	2079	2114	2149
30	52	+75, -15	23	120	80	20	34.5	71	G2	2011	2046	2081	2116	2151
30	150	+75, -15	60	360	240	20	34.5	25	G3	2012	2047	2082	2117	2152
30	320	+75, -15	125	750	500	20	34.5	11.5	G4	2013	2048	2083	2118	2153
30	460	+75, -15	165	1000	700	20	34.5	8	G5	2014	2049	2084	2119	2154
50	25	+50, -15	19	100	60	30	57.5	150	G2	2016	2051	2086	2121	2156
50	75	+50, -15	50	300	180	30	57.5	49	G3	2017	2052	2087	2122	2157
50	160	+50, -15	100	640	390	30	57.5	23	G4	2018	2053	2088	2123	2158
50	230	+50, -15	150	920	550	30	57.5	16	G5	2019	2054	2089	2124	2159
75	15	+50, -15	17	90	60	50	86.2	250	G2	2021	2056	2091	2126	2161
75	45	+50, -15	45	270	180	50	86.2	83	G3	2022	2057	2092	2127	2162
75	95	+50, -15	95	570	380	50	86.2	39	G4	2023	2058	2093	2128	2163
75	130	+50, -15	130	780	520	50	86.2	28	G5	2024	2059	2094	2129	2164
100	10	+30, -15	15	80	52	65	115.0	370	G2	2026	2061	2096	2131	2166
100	30	+30, -15	40	230	150	65	115.0	121	G3	2027	2062	2097	2132	2167
100	63	+30, -15	80	480	310	65	115.0	58	G4	2028	2063	2098	2133	2168
100	90	+30, -15	120	720	470	65	115.0	41	G5	2029	2064	2099	2134	2169
150	5.5	+30, -15	12	64	42	100	172.0	675	G2	2031	2066	2101	2136	2171
150	17	+30, -15	30	190	130	100	172.0	220	G3	2032	2067	2102	2137	2172
150	35	+30, -15	65	400	260	100	172 0	106	G4	2033	2068	2103	2138	2173
150	50	+30, -15	95	560	380	100	172.0	74	G5	2034	2069	2104	2139	2174

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T16:16Z
Check the source to verify that this is the current version before use.

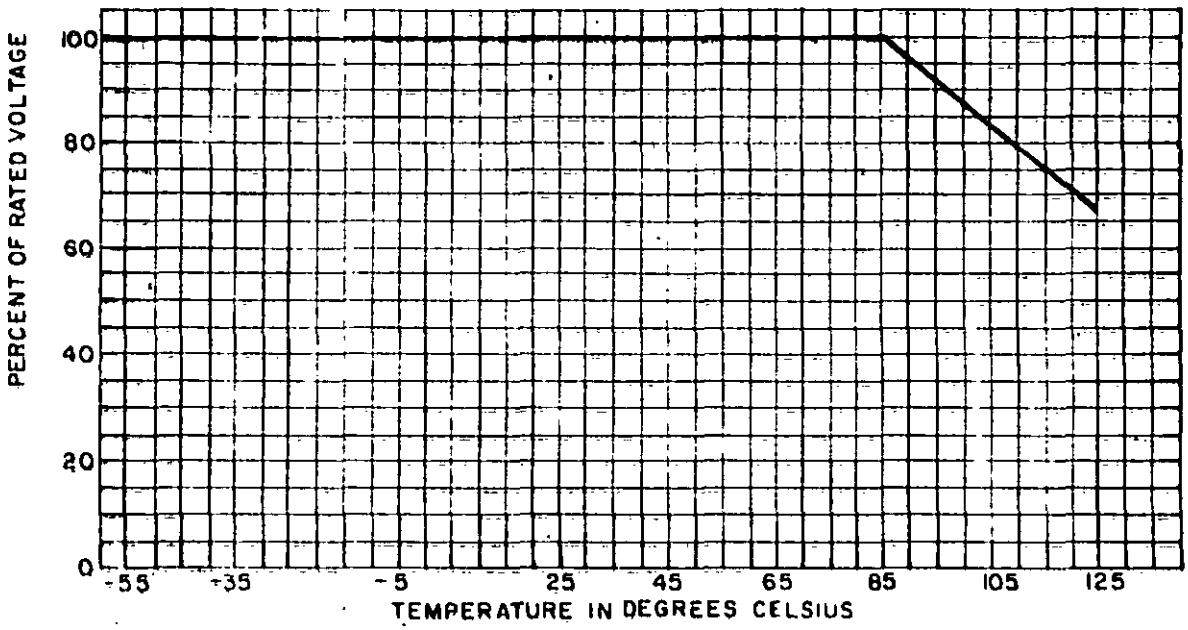


FIGURE 2. Voltage derating with temperature.

Cap. (nom): Method 305 of MIL-STD-202. See table I.

Dissipation factor (DF) (max):
50 volts and below - 20%.
Above 50 volts - 15%.

Seal: (G seal only): Method 112 of MIL-STD-202, conditions A and C.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I (100G).

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20G).

Temperature cycling: Method 102 of MIL-STD-202, condition D (with step 3 at +125°C). Number of cycles - 30.
DCL - 125% (max) of 25°C value of table I.
ΔCap. - Within ±5% of initial measurement.
DF - 115% (max) of initial requirement.

Salt spray (corrosion): Method 101 of MIL-STD-202, condition B (48 hr).

Solderability: Method 208 of MIL-STD-202.

Terminal strength:
Pull - Method 211 of MIL-STD-202, condition A.
Wire-lead bend - In accordance with MIL-C-39006.

Surge voltage: See table I.

Moisture resistance: Method 106 of MIL-STD-202.
DCL - 125% (max) of 25°C value of table I.
ΔCap. - Within ±6% of initial measurement.
DF - 115% (max) of initial requirement.

Dielectric withstanding voltage (DWV): Method 301 of MIL-STD-202. 2,000 Vdc, min.

Insulation resistance (IR): Method 302 of MIL-STD-202, condition B. 100 megohms, min.

Low temperature (storage): Method 502 of MIL-STD-810.
DCL - See table I.
ΔCap. - Within ±5% of initial measurement.
DF - 20% (≤50V) or 15% (>50V) (max).

Stability at low and high temperatures:
Step 1 (+25°C):
DCL - See table I.
Cap. - Within tolerance of table I value.
DF - 20% (≤50V) or 15% (>50V) (max).
Step 2 (-55°C):
Imp. - See table I.
ΔCap. - -40% of step 1 value (max).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:16Z
Check the source to verify that this is the current version before use.

Step 3 (+25°C):
DCL - See table I.
ΔCap. - Within ±5% of step 1 value.
DF - 20% (≤50V) or 15% (>50V) (max).
Step 4 (+85°C):
DCL - See table I.
ΔCap. - +50% of step 1 value (max).
DF - 55% (≤50V) or 45% (>50V) (max).
1/ Step 5 (+125°C):
DCL - See table I.
ΔCap. - +75% of step 1 value (max).
DF - 55% (≤30V) or 45% (>30V) (max).
Step 6 (+25°C):
DCL - See table I.
ΔCap. - Within ±5% of step 1 value.
DF - 20% (≤50V) or 15% (>50V) (max).

Reverse voltage:
DCL - 125% (max) of 25°C value of table I.
ΔCap. - Within ±10% of initial measurement
DF - 20% (≤50V) or 15% (>50V) (max).

Life at 85°C: Method 108 of MIL-STD-202.
2,000-hr (qualification):
DCL - See table I.
ΔCap. - Within ±15% of initial measurement.
DF - 20% (≤50V) or 15% (>50V) (max).
240-hr (group B):
DCL - See table I.
ΔCap. - Within ±10% of initial measurement.
DF - 20% (≤50V) or 15% (>50V) (max).
10,000-hr (group C) and extended life:
DCL (85°C) - 125% (max) of table I value.
DCL (25°C) - See table I.
ΔCap. - Within ±25% of initial measurement.
DF - 40% (≤50V) or 30% (>50V) (max).
DWV - 2,000 vdc, min.
IR - 100 megohms, min.

1/ 125°C derated voltage referenced.

Ⓐ Vacuum life: (G seal only):
DCL - See table I.
ΔCap. - Within ±20% of initial measurement.
DF - 10% (max).
Weight loss - 4.1 milligrams, max.

1/ Life at 125°C: Method 108 of MIL-STD-202.
DCL -See table I.
ΔCap - Within ±15% of initial measurement.
DF - 20% (≤30V) or 15% (>30V) (max).
DWV - 2,000 vdc, min.
IR - 100 megohms, min.

Barometric pressure: Method 105 of MIL-STD-202, condition E (150,000 ft).
Test voltage - Rated voltage.

Fungus: Method 508 of MIL-STD-810.

Resistance to solvents: Method 215 of MIL-STD-202.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B.
DCL - See table I.
ΔCap. - Within ±5% of initial measurement.
DF - 20% (≤50V) or 15% (>50V) (max).

Marking: In accordance with MIL-C-39006.
Polarity - Series of plus (+) symbols encircling positive end (see figure 1).

Voltage group (85°C rated voltage):

I - - - - - - - 15 to 150V incl

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU
Navy - SH
Air Force - 11, 17, 80
DSA - ES

User activities:
Army -
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5910-1201)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:16Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT
DOD 316

OFFICIAL BUSINESS

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:16Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 202640-21T16:16Z
Check the source to verify that this is the current version before use.